COMPARING CLASSICAL AND PATH COUPLING BOUNDS IN FINITE-STATE MARKOV CHAINS: A GRAPH-COLORING CASE STUDY

IZ TECÚM

ABSTRACT. Bounding the convergence rate of Markov-chain Monte-Carlo (MCMC) samplers is central to probabilistic computation. Two popular techniques are (i) classical coupling, which constructs a joint process that coalesces, and (ii) path coupling, which verifies a one-step contraction in a chosen metric and lifts it to the whole state space. Quantitative head-to-head comparisons on the same chain are rare. This work studies single-site Glauber dynamics for q-colourings of the four-vertex cycle C_4 ($q \geq 3$). We provide

$$\mathbf{E}[\tau_{\mathrm{cl}}] \leq 6(q-1), \qquad \mathbf{E}[\tau_{\mathrm{pc}}] \leq 2(q-1),$$

where the latter follows from a Hamming path-coupling contraction $\alpha(q) = 1 - \frac{1}{2(q-1)}$. Consequently $t_{\mathrm{mix}}^{\mathrm{pc}}(\varepsilon) < t_{\mathrm{mix}}^{\mathrm{cl}}(\varepsilon)$ for all $q \geq 3$ and $\varepsilon \in (0, 1/2]$. Exact enumeration and 10^3 Monte-Carlo trajectories per start state corroborate the linear dependence on $(q-1)$ and display a constant-factor gap consistent with theory. The four-cycle thus serves as a transparent test-bed showing when path coupling outperforms classical coupling and suggesting metric choices for larger colouring chains.

DEPARTMENT OF MATHEMATICS, COLUMBIA UNIVERSITY, NEW YORK, NY 10027, USA
Email address: ilt2109@columbia.edu

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